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# LES-Based Estimation of Monopile-Induced Turbulence Source Parameters for Ocean Models

Post-Processing Report

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**Date** June 25 – July 8, 2026

**Simulation** LES of flow past a circular cylinder (OpenFOAM)

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# 1. Simulation Parameters

The following parameters describe the LES case used for all calculations in this report.

| Parameter                                    | Value                                      |
|----------------------------------------------|--------------------------------------------|
| Cylinder diameter D                          | 8 m                                        |
| Water depth H                                | 40 m                                       |
| Inflow velocity $U_0$                        | 0.5 m/s                                    |
| Reynolds number Re                           | $4 \times 10^6$                            |
| Domain size                                  | 600 × 120 × 40 m                           |
| Cylinder center                              | (40, 60, 0) m                              |
| Averaging period                             | 157 s                                      |
| Sampling interval                            | 1 s                                        |
| Reference density $\rho_0$                   | 1025 kg/m <sup>3</sup>                     |
| Control volume $\Delta X \times \Delta Y$    | 300 m × 140 m (first stage)                |
| $A_{\text{cell}} = \Delta X \times \Delta Y$ | 42,000 m <sup>2</sup>                      |
| $V_c = A_{\text{cell}} \times H$             | $42,000 \times 40 = 1,680,000 \text{ m}^3$ |

## 2. Resolved TKE Maps (Outcome 1)

This section reports the resolved turbulent kinetic energy ( $k^{\text{Res}}$ ) extracted from the LES data.

### Step 1: Load LES data

Load the three-dimensional fields  $u(x,y,z,t^n)$ ,  $v(x,y,z,t^n)$ ,  $w(x,y,z,t^n)$ , and  $p(x,y,z,t^n)$  at all saved time steps. Also load grid information including cell volumes and grid spacing.

#### **In brief:**

Open ParaView and load case\_uniform.foam. Hit Apply.

### Step 2: Define geometry, masks, and control volume

Define the cylinder geometry and exclude solid cells from all volume averages. Define control volume  $V_c = A_{\text{CEL}}H$ , where  $A_{\text{CEL}} = \Delta X \Delta Y$ . At the first stage, choose  $\Delta X = 300 \text{ m}$  and  $\Delta Y = 140 \text{ m}$ .

#### **Result:**

$\Delta X = 300 \text{ m}$ ,  $\Delta Y = 140 \text{ m}$ ,  $H = 40 \text{ m}$ ,  $A_{\text{cell}} = 42,000 \text{ m}^2$ ,  $V_c = 1,680,000 \text{ m}^3$ .

#### **In brief:**

Just calculate the numbers using math and the data given, no paraview action needed.

### Step 3: Calculate the local time-mean flow and fluctuations

Calculate the local time-mean velocity fields  $\bar{u}(x,y,z)$ ,  $\bar{v}(x,y,z)$ , and  $\bar{w}(x,y,z)$  using the statistically steady averaging period. Calculate turbulent fluctuations as  $u' = u - \bar{u}$ ,  $v' = v - \bar{v}$ , and  $w' = w - \bar{w}$ .

#### **In brief:**

- ExtractTimeSteps, edit the start and end time to around 64 time steps (3 shedding cycles) so it reduces the time to run the following filters, I chose  $t = 232$  to  $t = 264$
- TemporalStatistics. Check Compute Average only. Apply. This is TemporalStatistics1 gives  $U_{\text{average}}$ .
- Select extract time steps and TemporalStatistics1 together (Ctrl-click both in Pipeline Browser). Append Attributes. Apply.
- Check Spreadsheet view on Append Attributes output, confirm  $U_{\text{average}}$  is same order of magnitude as  $U$  (not off by  $\sim 100x$ ) before continuing.
- Calculator on the Append Attributes output. Result Array Name fluctuation, Result Array Type Vector. Expression:  $U - U_{\text{average}}$ . Apply.

### Step 4: Calculate resolved TKE while ignoring SGS kinetic energy

Compute  $k^{\text{Res}} = \frac{1}{2}(\bar{u'^2} + \bar{v'^2} + \bar{w'^2})$  at every grid point. SGS kinetic energy is neglected at this stage, so  $k^{\text{Res}} \approx k^{\text{Res}}$ .

#### **In brief:**

- Calculator on the fluctuation Calculator. Result Array Name  $u^2$ . Expression:  $\text{fluctuation}_X^2$  (check the array dropdown for the actual component suffix, X/Y/Z or 0/1/2). Apply.
- Calculator on that (chained). Result Array Name  $v^2$ . Expression:  $\text{fluctuation}_Y^2$ . Apply.
- Calculator on that (chained). Result Array Name  $w^2$ . Expression:  $\text{fluctuation}_Z^2$ . Apply.
- TemporalStatistics on that last Calculator. Check Compute Average only. Apply. This is TemporalStatistics2, gives  $u^2_{\text{average}}$ ,  $v^2_{\text{average}}$ ,  $w^2_{\text{average}}$ .
- Calculator on TemporalStatistics2. Result Array Name  $k_{\text{res}}$ . Expression:  $0.5*(u^2_{\text{average}}+v^2_{\text{average}}+w^2_{\text{average}})$ . Apply.
- Slice at  $z=0$  (horizontal): Origin: (40, 60, 0), Normal: (0, 0, 1)
- Slice at  $y=60$  (vertical): Origin: (40, 60, 0), Normal: (0, 1, 0)
- Color both by  $k_{\text{res}}$ , screenshot both for the maps.

| Field                                                    | Value                  | Unit                    |
|----------------------------------------------------------|------------------------|-------------------------|
| Integrated $k_{\text{res}}$ ( $\int k_{\text{res}} dV$ ) | 834.696                | $\text{m}^5/\text{s}^2$ |
| $V_c$ used                                               | 2,880,000              | $\text{m}^3$            |
| $(k_{\text{res}})_{Vc}$                                  | $2.898 \times 10^{-4}$ | $\text{m}^2/\text{s}^2$ |

## 2.1 Horizontal (x-y) plane slice

| Field                                   | Value                         | Unit                    |
|-----------------------------------------|-------------------------------|-------------------------|
| $k^{\text{Res}}$ range (min)            | 0                             | $\text{m}^2/\text{s}^2$ |
| $k^{\text{Res}}$ range (max)            | 1.25e-01                      | $\text{m}^2/\text{s}^2$ |
| Location of peak $k^{\text{Res}}$ (x/D) | $(54.923 - 40) / 8 = 1.865$   |                         |
| Location of peak $k^{\text{Res}}$ (y/D) | $(58.7522 - 60) / 8 = -0.156$ |                         |
| Reference                               | Horizontal TKE                |                         |

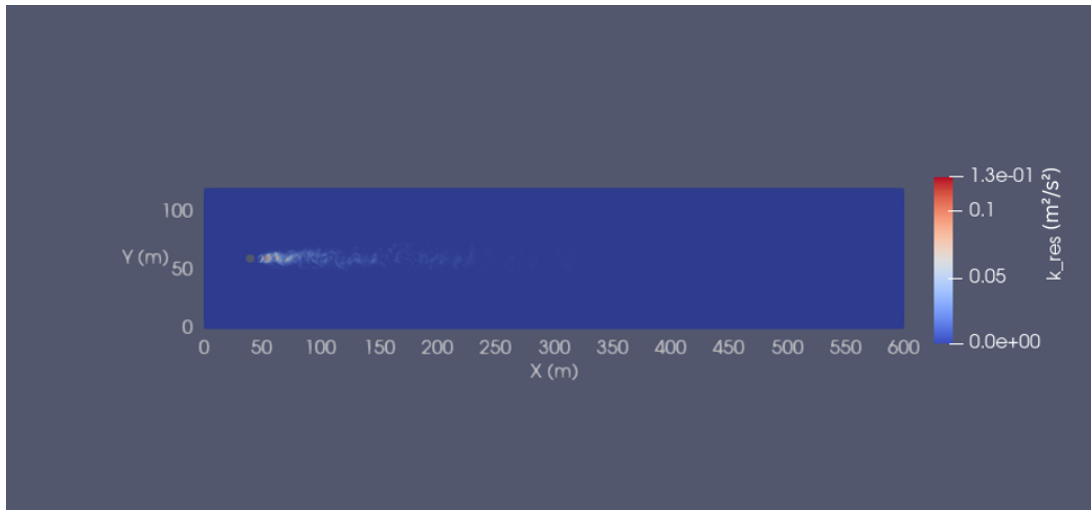


Figure 2.1 — Horizontal (x-y) plane slice of resolved TKE ( $k^{\text{Res}}$ ).

## 2.2 Vertical (x-z) plane slice through cylinder centreline

| Field                                 | Value        | Unit                    |
|---------------------------------------|--------------|-------------------------|
| $k^{\text{Res}}$ range observed (min) | 0            | $\text{m}^2/\text{s}^2$ |
| $k^{\text{Res}}$ range observed (max) | 2.19 e-01    | $\text{m}^2/\text{s}^2$ |
| Vertical extent of elevated k (z min) | -20          | m                       |
| Vertical extent of elevated k (z max) | 20           | m                       |
| Reference                             | Vertical TKE |                         |

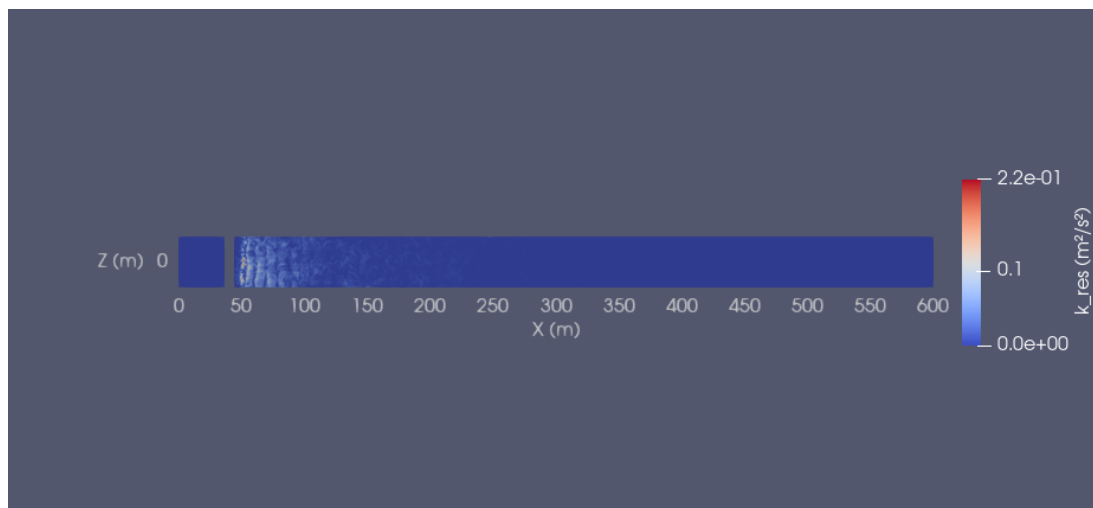


Figure 2.2 — Vertical (x-z) plane slice of resolved TKE ( $k^{Res}$ ).

### 3. Resolved Dissipation Maps (Outcome 2)

This section reports the resolved dissipation rate ( $\epsilon^{\text{Res}}$ ) computed from the fluctuating strain rate tensor.

#### Step 8: Estimate dissipation

$\epsilon_{\text{res}} = 2\nu s'_{ij} s'_{ij}$ , where  $s'_{ij}$  is the fluctuating strain rate tensor and  $\nu = 10^{-6} \text{ m}^2/\text{s}$ .

$\epsilon_{\text{SGS}} \approx 2\nu_{\text{SGS}} S_{ij} S_{ij}$ , where  $\nu_{\text{SGS}}$  is the SGS eddy viscosity ("nut") and  $S_{ij}$  is the strain rate tensor of the instantaneous resolved velocity field.

$\epsilon_{\text{LES}} = \epsilon_{\text{res}} + \epsilon_{\text{SGS}}$ . Use this total for calibrating  $c_4$ . If nut is unavailable, fall back to  $\epsilon_{\text{LES}} \approx \epsilon_{\text{res}}$ , noting this likely underestimates dissipation near the cylinder and in the near wake.

The following pipeline to solve  $\epsilon_{\text{LES}}$ :

- 8.0 (this step is to speed up the process, not necessary) ExtractTimeSteps, select time range, I used  $t = 232$  to  $t = 264$ . Make sure mesh region is internal mesh and "nut" is selected along p and U in the properties.
- 8.1 Append Attributes on ExtractTimeSteps + TemporalStatistics1 (only compute average). Apply. This gives  $U_{\text{average}}$  for the fluctuation calculation in 8.2.
- 8.2 Calculator: fluctuation (Vector) =  $U - U_{\text{average}}$ . Apply.
- 8.3 Gradient on fluctuation. Change scalar array to fluctuation, Compute Gradient. Apply.
- 8.4 Calculator:  $\text{eps\_res\_inst} = 0.000002 * (\text{Gradients\_0}^2 + \text{Gradients\_4}^2 + \text{Gradients\_8}^2 + 0.5 * (\text{Gradients\_1} + \text{Gradients\_3})^2 + 0.5 * (\text{Gradients\_2} + \text{Gradients\_6})^2 + 0.5 * (\text{Gradients\_5} + \text{Gradients\_7})^2)$ . Apply. This uses the constant molecular viscosity, not nut.
- 8.5 Upon appendattributes1, Gradient on U (the instantaneous velocity field, not fluctuation). Change scalar array to U, Compute Gradient. Apply. Name the output array Gradients\_U.
- 8.6 Calculator:  $\text{eps\_sgs\_inst} = 2 * \text{nut} * (\text{Gradients\_U\_0}^2 + \text{Gradients\_U\_4}^2 + \text{Gradients\_U\_8}^2 + 0.5 * (\text{Gradients\_U\_1} + \text{Gradients\_U\_3})^2 + 0.5 * (\text{Gradients\_U\_2} + \text{Gradients\_U\_6})^2 + 0.5 * (\text{Gradients\_U\_5} + \text{Gradients\_U\_7})^2)$ . Apply.
- 8.7 Append attributes on calculator eps\_sgs\_inst and eps\_res\_inst
- 8.8 Calculator:  $\text{eps\_LES\_inst} = \text{eps\_res\_inst} + \text{eps\_sgs\_inst}$ . Apply.
- 8.8 TemporalStatistics on eps\_LES\_inst. Compute Average only. Apply. This gives ( $\epsilon_{\text{LES}}$ ). Don't save data here yet, it's going to waste you hours.
- 8.10 Horizontal Slice ( $z = 0$ ). Filters → Slice. Set Origin to (40, 60, 0) and Normal to (0, 0, 1). Apply. Color by eps\_LES\_inst\_average. Make sure x and y axis shows.
- 8.10 Vertical Slice ( $y = 60$ ). Filters → Slice. Set Origin to (40, 60, 0) and Normal to (0, 1, 0). Apply. Color by eps\_LES\_inst\_average.
- 8.11 Integrate Variables on eps\_LES\_inst\_average (full 3D, no slice). Divide by  $V_c = 1,680,000 \rightarrow (\epsilon_{\text{LES}})$ . Fills table 8.1
- 8.12 Append Attributes:  $k_{\text{res}} + \text{eps\_LES\_inst\_average}$ . Threshold  $k_{\text{res}} > \sim 1e-6$ . Calculator  $\text{eps2\_over\_k} = \text{eps\_LES\_inst\_average}^2 / k_{\text{res}}$ . Calculator  $\text{eps\_over\_k\_Sk} = \text{eps\_LES\_inst\_average} / k_{\text{res}}$ .
- 8.13 Integrate Variables on eps2\_over\_k → table 8.2 row1; Integrate Variables on eps\_over\_k  $\times 2.153e-6 \rightarrow$  table 8.2 row2;  $c_4 = 1.92 \times \text{row1} / \text{row2} \rightarrow$  table 8.2 row3.

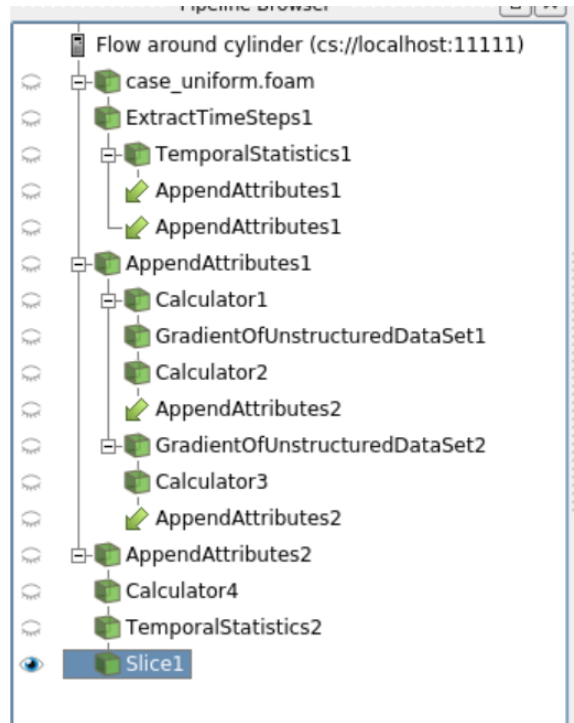
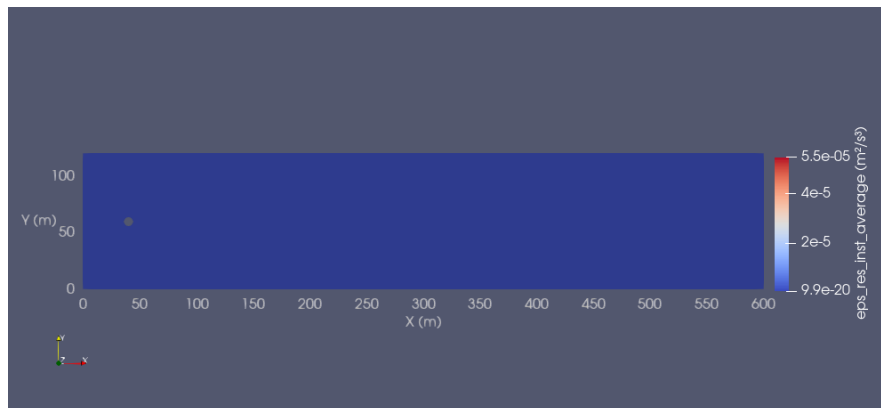


Figure 3.0 — ParaView pipeline browser for the dissipation-rate workflow.

### 3.1.1 Horizontal plane slice of $\epsilon^{\text{res}}$

| Field                                        | Value                                        | Unit                    |
|----------------------------------------------|----------------------------------------------|-------------------------|
| $\epsilon^{\text{res}}$ range observed (min) | 9.9e-20                                      | $\text{m}^2/\text{s}^3$ |
| $\epsilon^{\text{res}}$ range observed (max) | 5.5e-05                                      | $\text{m}^2/\text{s}^3$ |
| D (diameter)                                 | 8                                            | m                       |
| X Location of peak $\epsilon$ (x/D)          | $(41.4725 - 40) / 8 = 1.4725 / 8 = 0.1841$   |                         |
| Y Location of peak $\epsilon$ (y/D)          | $(56.2809 - 60) / 8 = -3.7191 / 8 = -0.4649$ |                         |
| Data Reference                               | Link                                         |                         |



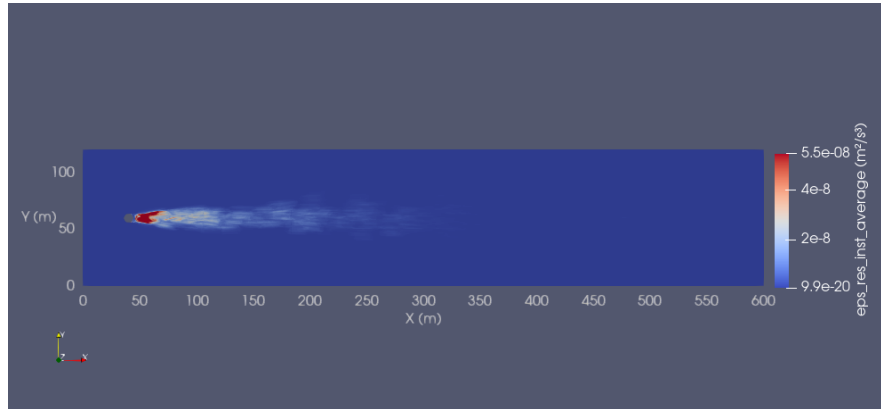
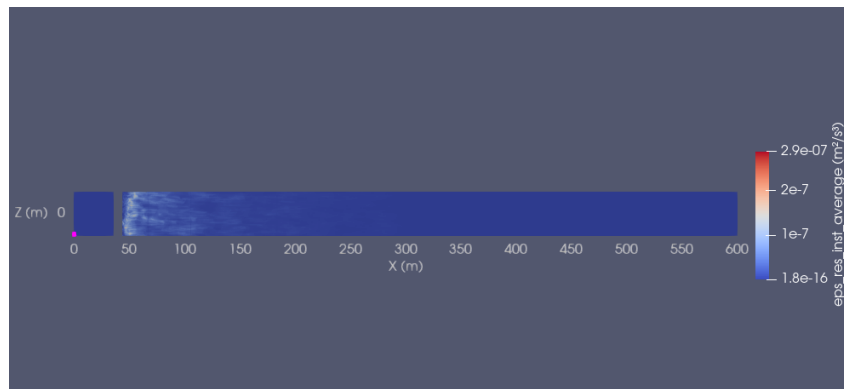
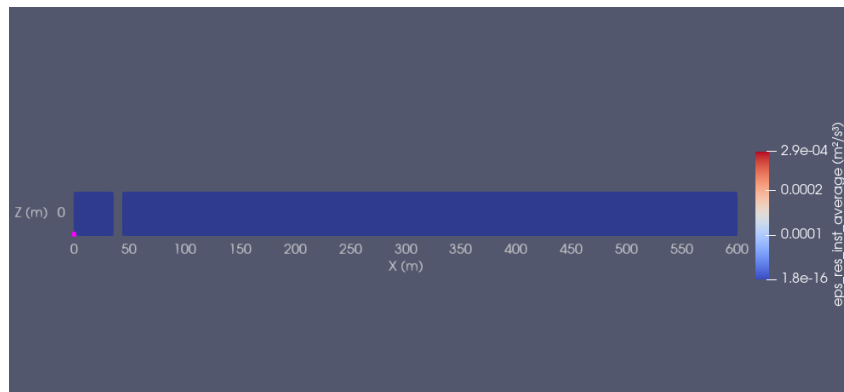


Figure 3.1.1.1 — Horizontal plane slice of resolved dissipation ( $\epsilon^{\text{res}}$ ).

### 3.1.2 Vertical plane slice of $\epsilon^{\text{res}}$

| Field                                        | Value                | Unit                           |
|----------------------------------------------|----------------------|--------------------------------|
| $\epsilon^{\text{res}}$ range observed (min) | 1.8e-16              | m <sup>2</sup> /s <sup>3</sup> |
| $\epsilon^{\text{res}}$ range observed (max) | 2.9e-04              | m <sup>2</sup> /s <sup>3</sup> |
| D (diameter)                                 | 8                    | m                              |
| Data Reference                               | <a href="#">Link</a> |                                |



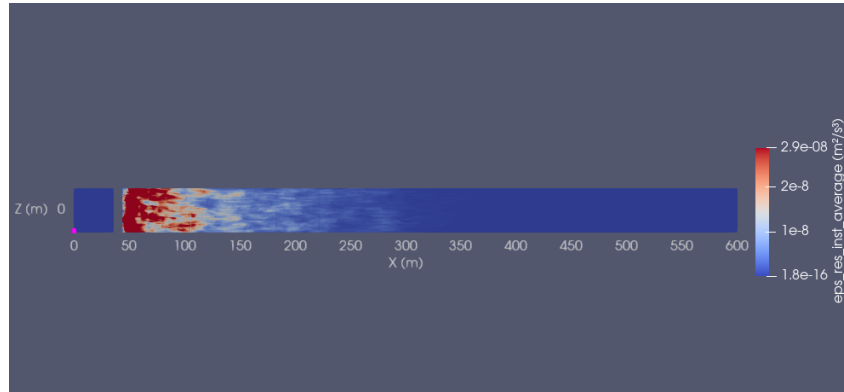
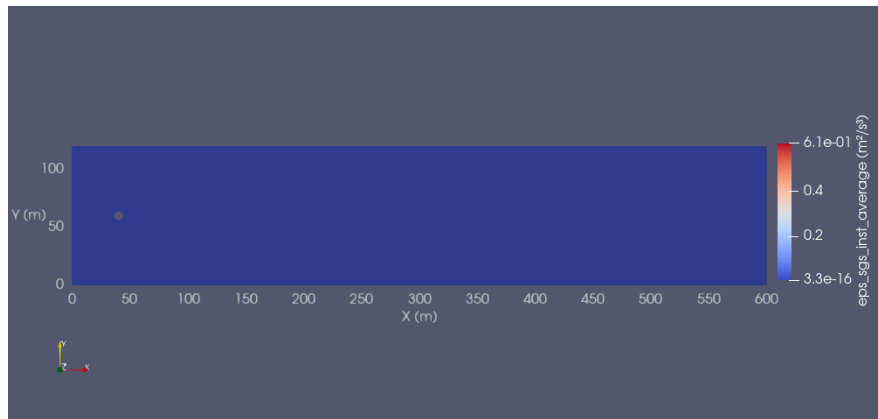


Figure 3.1.2 — Vertical plane slice of resolved dissipation ( $\epsilon^{res}$ ).

### 3.2.1 Horizontal plane slice of $\epsilon^{sGs}$

| Field                                 | Value                           | Unit      |
|---------------------------------------|---------------------------------|-----------|
| $\epsilon^{sGs}$ range observed (min) | 3.3e-16                         | $m^2/s^3$ |
| $\epsilon^{sGs}$ range observed (max) | 6.1e-01                         | $m^2/s^3$ |
| D (diameter)                          | 8                               | m         |
| X Location of peak $\epsilon$ (x/D)   | $(39.0043 - 40) / 8 = -0.12446$ |           |
| Y Location of peak $\epsilon$ (y/D)   | $(56.1219 - 60) / 8 = -0.48476$ |           |
| Data Reference                        | <a href="#">Link</a>            |           |



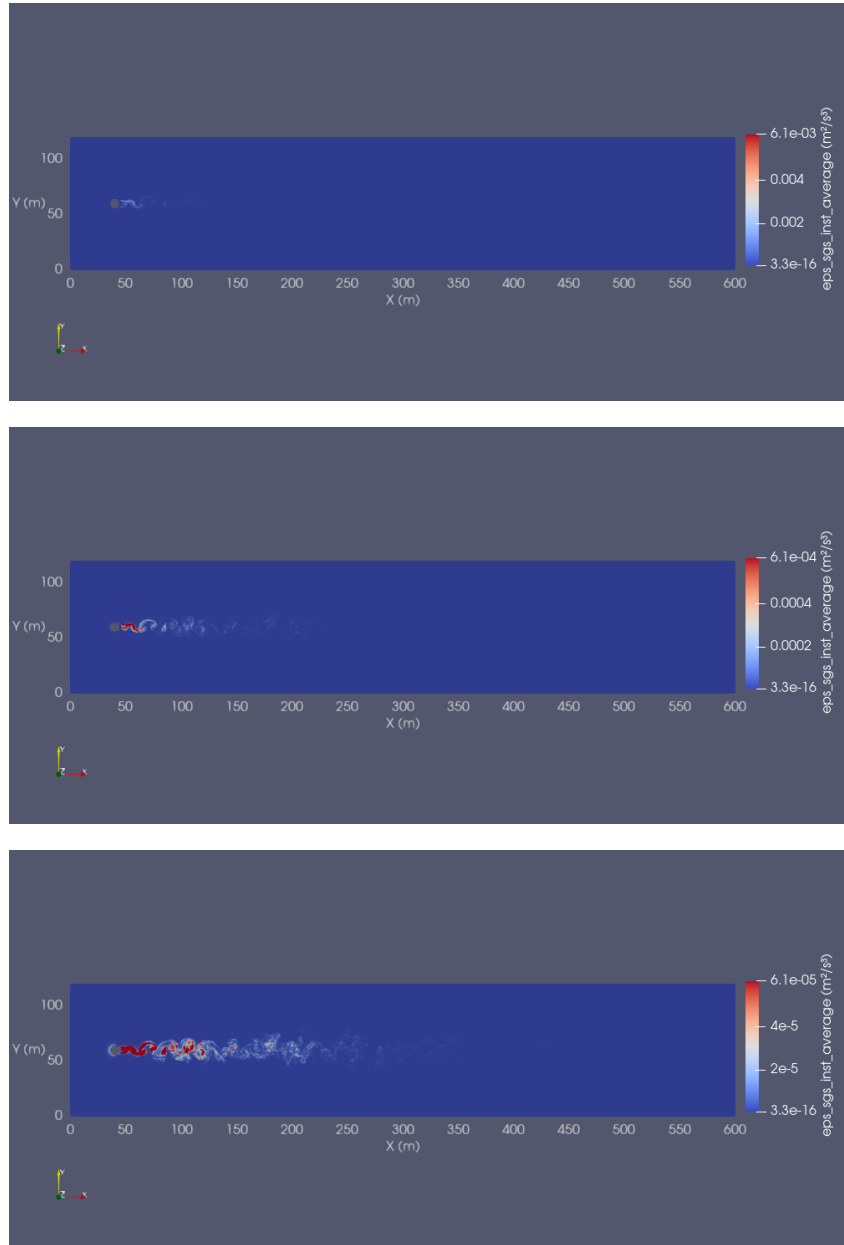


Figure 3.2.1 — Horizontal plane slice of SGS dissipation ( $\epsilon^{SGs}$ ).

### 3.2.2 Vertical plane slice of $\epsilon^{SGs}$

| Field                                 | Value   | Unit                           |
|---------------------------------------|---------|--------------------------------|
| $\epsilon^{SGs}$ range observed (min) | 5.3e-13 | m <sup>2</sup> /s <sup>3</sup> |
| $\epsilon^{SGs}$ range observed (max) | 1.1e-02 | m <sup>2</sup> /s <sup>3</sup> |
| D (diameter)                          | 8       | m                              |
| Data Reference                        | Link    |                                |

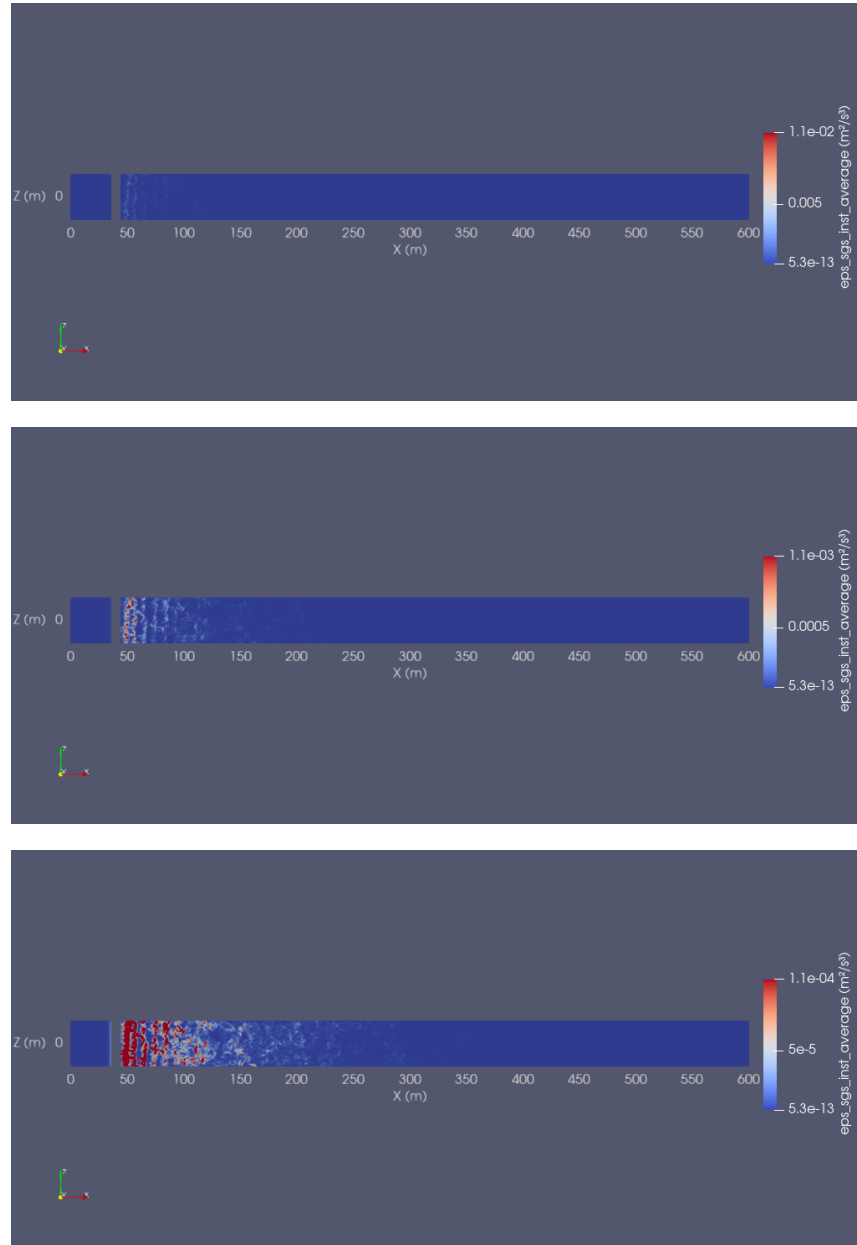


Figure 3.2.2 — Vertical plane slice of SGS dissipation ( $\epsilon^{SGS}$ ).

### 3.3.1 Horizontal plane slice of $\epsilon^{LES}$

| Field                                 | Value                          | Unit      |
|---------------------------------------|--------------------------------|-----------|
| $\epsilon^{LES}$ range observed (min) | 3.4e-16                        | $m^2/s^3$ |
| $\epsilon^{LES}$ range observed (max) | 0.61                           | $m^2/s^3$ |
| D (diameter)                          | 8                              | m         |
| X Location of peak $\epsilon$ (x/D)   | $(39.0043 - 40) / 8 = -0.1245$ |           |
| Y Location of peak $\epsilon$ (y/D)   | $(56.1219 - 60) / 8 = -0.4848$ |           |
| Data Reference                        | Horizontal Dissipation         |           |

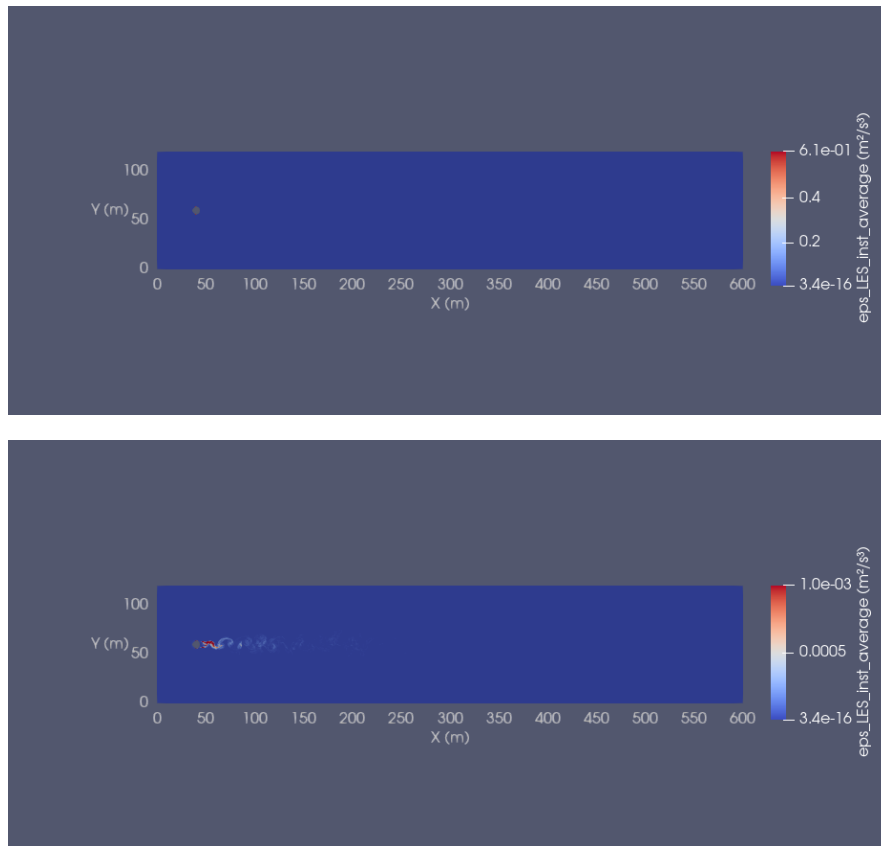
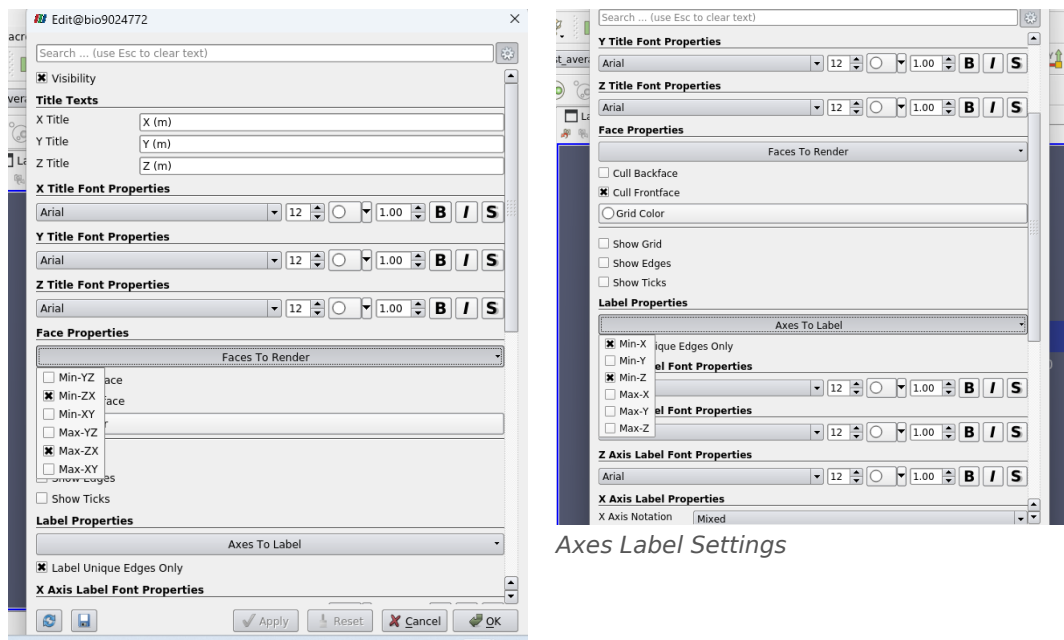


Figure 3.3.1 — Horizontal plane slice of total LES dissipation ( $\epsilon^{LES}$ ).

### 3.3.2 Vertical plane slice of $\epsilon^{LES}$

| Field                                 | Value                | Unit      |
|---------------------------------------|----------------------|-----------|
| $\epsilon^{LES}$ range observed (min) | 5.3e-13              | $m^2/s^3$ |
| $\epsilon^{LES}$ range observed (max) | 0.011                | $m^2/s^3$ |
| Data reference                        | Vertical Dissipation |           |



Axes Label Settings

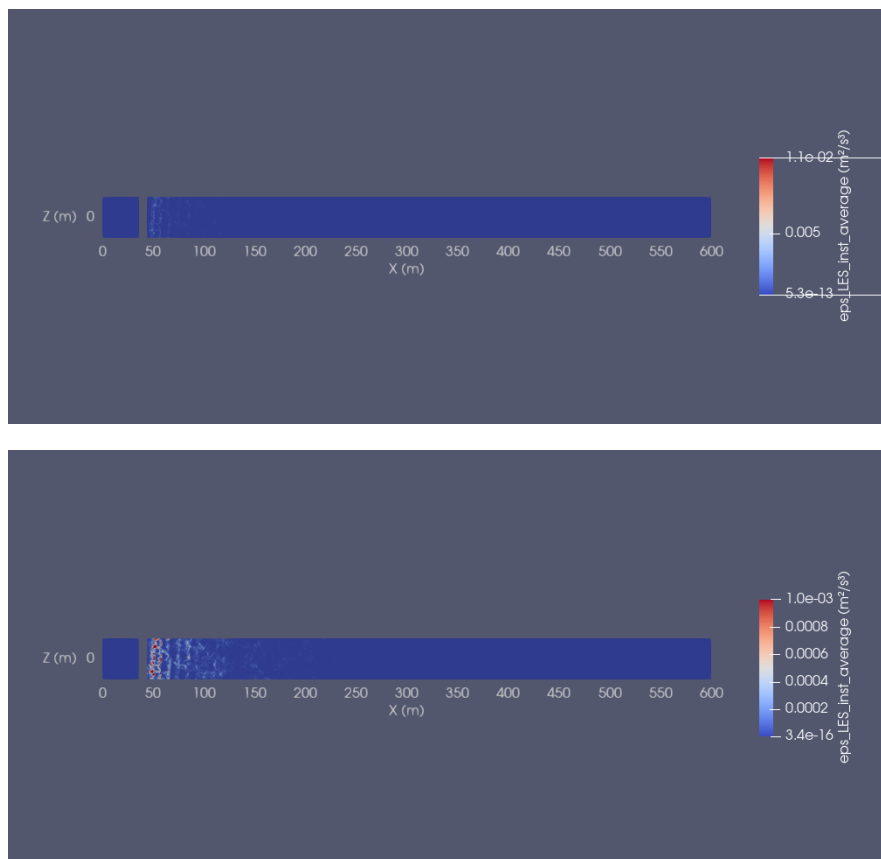


Figure 3.3.2 — Vertical plane slice of total LES dissipation ( $\epsilon^{LES}$ ).

## 4. Wake-Generated TKE Production Maps (Outcome 3)

This section reports wake-generated TKE production ( $P^{\text{Wake}}$ ) and the volume-averaged TKE source term.

### Step 5: Calculate Reynolds stresses

Using the local time-mean fluctuations from Step 3, compute the resolved Reynolds-stress components:  $u'u'$ ,  $u'v'$ ,  $u'w'$ ,  $v'u'$ ,  $v'v'$ ,  $v'w'$ ,  $w'u'$ ,  $w'v'$ , and  $w'w'$ . These are required for the TKE production calculation.

#### *In paraview:*

- Extract Time steps, here I used  $t = 232$  to  $t = 264$
- 5.1 TemporalStatistics on case\_uniform.foam. Check Compute Average and Compute Standard Deviation. Apply. This is TemporalStatistics1.
- 5.2 Calculator on TemporalStatistics1. Result Array Name  $u'u'$ . Expression  $U\_stddev\_0^2$ . Apply. This is Calculator1.
- 5.3 Calculator on Calculator1. Result Array Name  $v'v'$ . Expression  $U\_stddev\_1^2$ . Apply. This is Calculator2.
- 5.4 Calculator on Calculator2. Result Array Name  $w'w'$ . Expression  $U\_stddev\_2^2$ . Apply. This is Calculator3, now holding  $u'u'$ ,  $v'v'$ ,  $w'w'$ , plus  $U\_average$  and  $U\_stddev$ .
- 5.5 Select Extract time steps and Calculator3 together (Ctrl-click both in Pipeline Browser), Append Attributes. Apply. This merges the raw instantaneous U with  $U\_average$  and the diagonal stresses.
- 5.6 Calculator on the Append Attributes output. Result Array Name fluctuation, set Result Array Type to Vector. Expression:  $U - U\_average$
- 5.7 Calculator on Calculator4. Result Array Name  $u'v'$ . Expression  $fluctuation\_X * fluctuation\_Y$  (adjust suffix if needed). Apply. This is Calculator5.
- 5.8 Calculator on Calculator5. Result Array Name  $u'w'$ . Expression  $fluctuation\_X * fluctuation\_Z$ . Apply. This is Calculator6.
- 5.9 Calculator on Calculator6. Result Array Name  $v'w'$ . Expression  $fluctuation\_Y * fluctuation\_Z$ . Apply. This is Calculator7, now holding  $u'v'$ ,  $u'w'$ ,  $v'w'$  together.
- 5.10 TemporalStatistics on Calculator7. Check Compute Average only, uncheck the rest. Apply. This is TemporalStatistics2.

### Step 6: Calculate mean velocity gradients

Using the local time-mean LES flow from Step 3, calculate the full velocity-gradient tensor  $\partial \bar{u}_i / \partial x_j$ . This includes all gradients of  $\bar{u}$ ,  $\bar{v}$ , and  $\bar{w}$  in the  $x$ ,  $y$ , and  $z$  directions. The monopile wake has strong local gradients that drive wake-generated TKE production.

#### *In paraview:*

- 6.1 Gradient filter on TemporalStatistics1 (the branch holding  $U\_average$ ). Input Array =  $U\_average$ , check Compute Gradient only. Apply. This is Gradient1, producing the 9-component tensor ( $0 = \partial u / \partial x$ ,  $1 = \partial u / \partial y$ ,  $2 = \partial u / \partial z$ ,  $3 = \partial v / \partial x$ ,  $4 = \partial v / \partial y$ ,  $5 = \partial v / \partial z$ ,  $6 = \partial w / \partial x$ ,  $7 = \partial w / \partial y$ ,  $8 = \partial w / \partial z$ ).

## Step 7: Calculate wake-generated TKE production

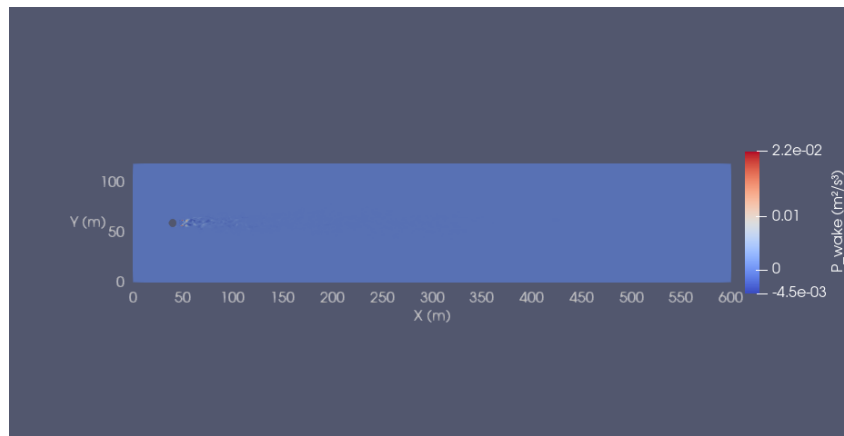
Compute  $P^{Wake} = -u'iu'j (\partial \bar{u}i / \partial xj)$  at every grid point. This must use the local time-mean wake flow, not the uniform inlet velocity (which has no spatial gradients). Then coarse-grain over the control volume:  $S_{k,LES}^{s3es} = (1/Vc) \int P^{Wake} dV$ .

### In paraview:

- Select TemporalStatistics2 and Gradient1 together, Append Attributes. Apply. This is AppendAttributes2.
- Select Calculator3 and AppendAttributes2 together, Append Attributes. Apply. This is AppendAttributes3, now holding all Reynolds stresses and velocity gradients.
- Calculator on AppendAttributes3. Result Array Name P\_wake. Expression  $-(u'u'*Gradients\_0 + u'v'\_average*Gradients\_1 + u'w'\_average*Gradients\_2 + u'v'\_average*Gradients\_3 + v'v'*Gradients\_4 + v'w'\_average*Gradients\_5 + u'w'\_average*Gradients\_6 + v'w'\_average*Gradients\_7 + w'w'*Gradients\_8)$ . Apply.
- Slice on P\_wake Calculator, Origin (40,60,0), Normal (0,0,1). Color by P\_wake, record min/max.
- Slice on P\_wake Calculator, Origin (40,60,0), Normal (0,1,0). Color by P\_wake, record min/max.
- Integrate Variables on P\_wake Calculator, divide result by  $Vc = 1,680,000 \text{ m}^3$  to get  $S_{k,LES}$ .

### 4.1.1 Horizontal plane slice of P\_wake

| Field                               | Value                                       | Unit                    |
|-------------------------------------|---------------------------------------------|-------------------------|
| P_wake range observed (min)         | -4.5e-03                                    | $\text{m}^2/\text{s}^3$ |
| P_wake range observed (max)         | 2.2e-02                                     | $\text{m}^2/\text{s}^3$ |
| X Location of peak $\epsilon$ (x/D) | $(50.0225 - 40) / 8 = 10.0225 / 8 = 1.2528$ |                         |
| Y Location of peak $\epsilon$ (y/D) | $(62.7075 - 60) / 8 = 2.7075 / 8 = 0.338$   |                         |
| Data Reference                      | PwakeHorizontal.xlsx                        |                         |



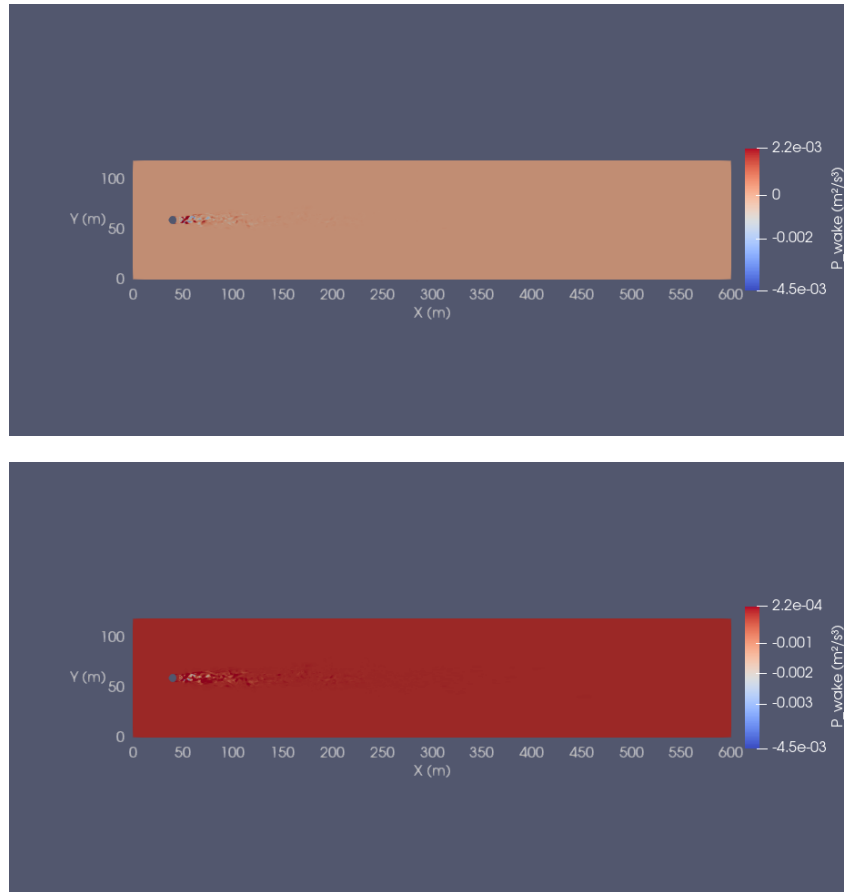
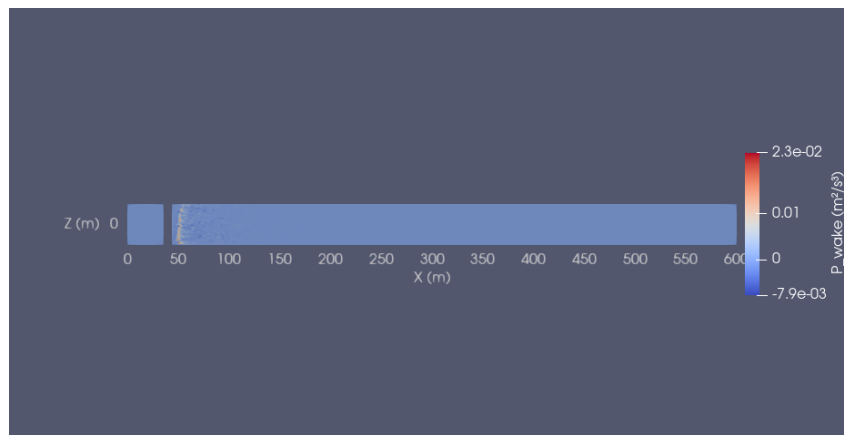


Figure 4.1.1 — Horizontal plane slice of wake-generated TKE production ( $P^{Wake}$ ).

#### 4.1.2 Vertical plane slice of $P_{wake}$

| Field                           | Value              | Unit      |
|---------------------------------|--------------------|-----------|
| $P_{wake}$ range observed (min) | -7.9e-03           | $m^2/s^3$ |
| $P_{wake}$ range observed (max) | 2.3e-02            | $m^2/s^3$ |
| Data reference                  | PwakeVertical.xlsx |           |



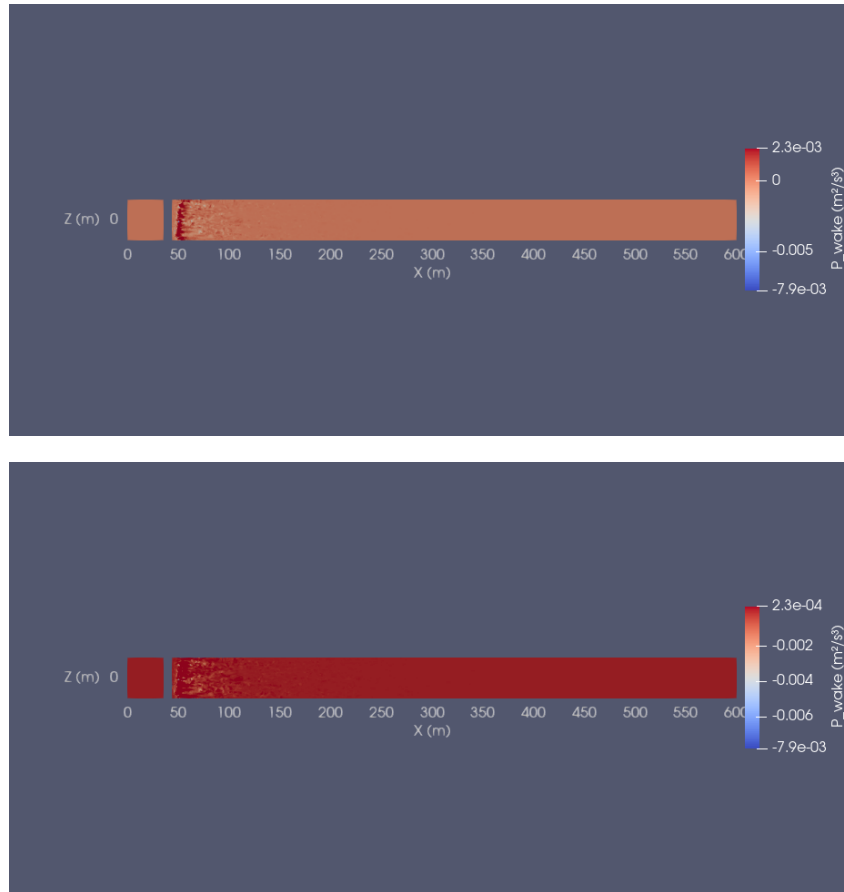


Figure 4.1.2 — Vertical plane slice of wake-generated TKE production ( $P^{Wake}$ ).

## 4.2 Control volume average

| Field                                                       | Value                              | Unit                           |
|-------------------------------------------------------------|------------------------------------|--------------------------------|
| Control volume used ( $\Delta X \times \Delta Y \times H$ ) | 600 m $\times$ 120 m $\times$ 40 m |                                |
| Integrated Pwake                                            | 6.20158                            | m <sup>5</sup> /s <sup>3</sup> |
| $S_{k_{LES}}^k$ ,LES (volume-averaged $P^{Wake}$ )          | $2.153 \times 10^{-6}$             | m <sup>2</sup> /s <sup>3</sup> |

## 5. Effective Drag Coefficient $C_{D,eff}$ (Outcome 4)

This section reports the effective drag coefficient estimated from the time-averaged drag force on the cylinder.

*Data source*

### Step 9: Calculate drag force and drag power

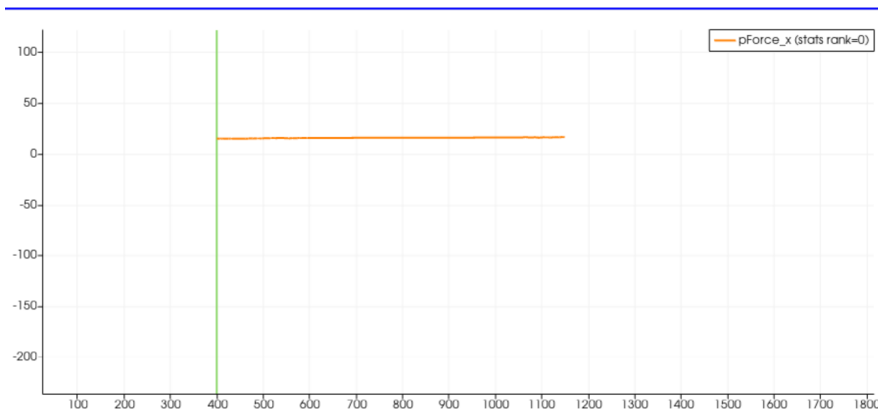
Calculate the drag force time series  $F_D(t)$  from the cylinder surface. Compute the time average  $\overline{F_D}$ . The effective drag coefficient is  $C_{D,eff} = \overline{F_D} / (\frac{1}{2} \rho_0 U_0^2 D H)$ , where  $H = 40$  m and  $U_0 = 0.5$  m/s. The total drag power is  $P_D^{Rak} = \overline{F_D} U_0$ .

#### Pipeline:

Reader (cylinder only) → Extract Surface → Generate Surface Normals → Calculator ( $pForce\_x = p * Normals\_X$ ) → Integrate Variables (check) → Plot Data Over Time

#### Steps after:

- 1 Select IntegrateVariables1 → Apply Plot Data Over Time
- 2 Switch view to Spreadsheet View
- 3 Set Attribute to Row Data (or Point Data if shown), find  $pForce\_x$  column across all timesteps
- 4 File → Save Data → export as CSV
- 5 In CSV/Excel/Python: average  $\overline{F_D}$  over steady-state window →  $\overline{F_D}$
- 6  $C_{D,eff} = \overline{F_D} / (0.5 \times \rho_0 \times U_0^2 \times D \times H)$
- 7  $P_D = \overline{F_D} \times U_0$

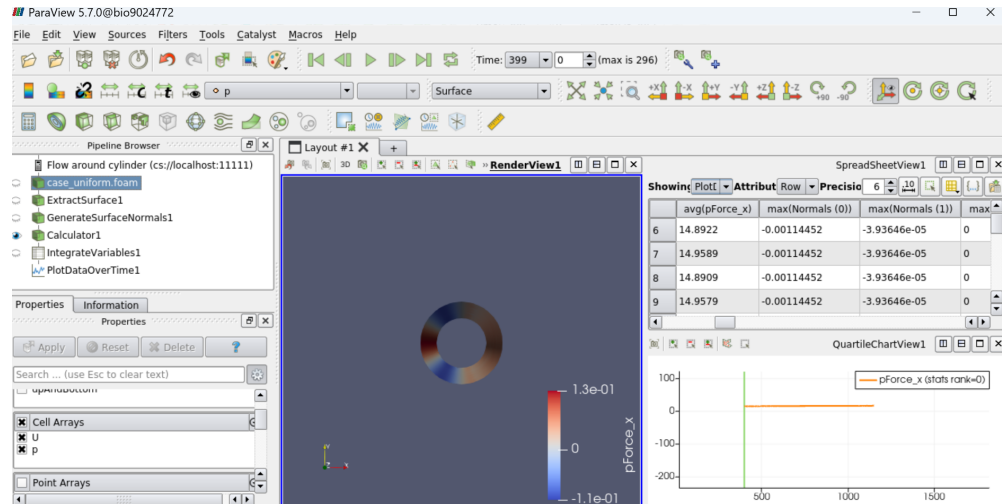


*x-axis represents time*

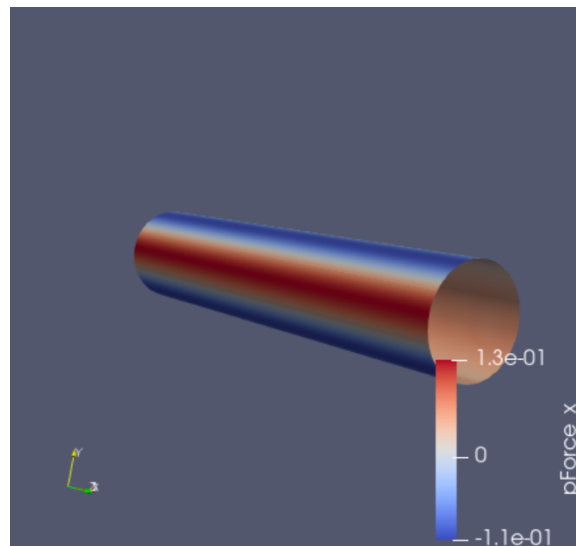
$$C_{D,eff} = \frac{\overline{F_D}}{\frac{1}{2} \rho_0 U_0^2 D H}$$

*\*the time-averaged  $F_D$  that Paraview generates is the kinematic drag force that already puts into consideration of  $p_0$  (already divided by  $p_0$ ), so ignore the  $p_0$  in the bottom\**

| Field                                      | Value       | Unit                                                      |
|--------------------------------------------|-------------|-----------------------------------------------------------|
| Time-averaged drag force $F_D$ (kinematic) | 16.17677    | $\text{m}^4/\text{s}^2 = \text{N}/(\text{kg}/\text{m}^3)$ |
| $C_{D,\text{eff}}$                         | 0.4044      |                                                           |
| Reference                                  | Data Source |                                                           |



Pipeline and graph.



Graph from a different angle.

## 6. Drag Power $P_D^{\text{Rak}}$ (Outcome 5)

This section reports the drag power per unit mass distributed over the ocean-model control volume.

The drag power per unit mass is:

$$P_{\text{drag}} = F_D \times U_0 / (\rho_0 \times A_{\text{cell}} \times H)$$

This should be consistent with the alternate form:

$$P_{\text{drag}} = \frac{1}{2} \times C_{D,\text{eff}} \times (D / A_{\text{cell}}) \times U_0^3$$

| Field                                        | Value                                                                                                                                                                                                                      | Unit                    |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| $A_{\text{cell}} = \Delta X \times \Delta Y$ | $300 \times 140 = 42,000 \text{ m}^2$                                                                                                                                                                                      | $\text{m}^2$            |
| Height                                       | 40                                                                                                                                                                                                                         | m                       |
| Total drag power ( $F_D \times U_0$ )        | 8,318.5                                                                                                                                                                                                                    | W                       |
| $P_D^{\text{Rak}}$                           | $4.83 \times 10^{-6} \text{ m}^2/\text{s}^3$                                                                                                                                                                               | $\text{m}^2/\text{s}^3$ |
| Consistency check with alternate form        | Both formulas agree exactly: $4.1 \times 10^{-6} \text{ m}^2/\text{s}^3$ from both $F_D \cdot U_0 / (\rho_0 \cdot A_{\text{cell}} \cdot H)$ and $\frac{1}{2} \cdot C_{D,\text{eff}} \cdot (D/A_{\text{cell}}) \cdot U_0^3$ |                         |

Computed from  $F_{D\_bar}$  using  $P_{\text{drag}} = F_{D\_bar} \times U_0 / (\rho_0 \times A_{\text{cell}} \times H)$ . Cross-checked with  $P_{\text{drag}} = 0.5 \times C_{D,\text{eff}} \cdot (D/A_{\text{cell}}) \cdot U_0^3$ . Both matched.

## 7. LES-Based Estimate of $\alpha^k$ (Outcome 6)

This section reports the parameter  $\alpha^k$ , which controls the fraction of drag power that becomes modelled TKE.

### Step 10: Estimate $\alpha^k$

Using  $S^{ksaes\_k,LES}$  from Step 7 and  $P_D^{Rak}$  from Step 9, compute  $\alpha^k = S^{ksaes\_k,LES} / P_D^{Rak}$  on the same control-volume basis. Also verify using the volume-integrated form:  $\alpha^k = \rho_0 \int P^{Wake} dV / (F_D U_0)$ . Repeat for multiple control-volume sizes to evaluate scale dependence.

### 7.1 Primary estimate ( $\Delta X = 600$ m, $\Delta Y = 120$ m)

| Field              | Value                         | Unit      |
|--------------------|-------------------------------|-----------|
| $S^{ksaes\_k,LES}$ | $2.15 \times 10^{-6}$         | $m^2/s^3$ |
| $P_D^{Rak}$        | $4.83 \times 10^{-6} m^2/s^3$ | $m^2/s^3$ |
| $\alpha^k$         | 0.445                         |           |

## 8. First Estimates of $c_4$ (Outcome 7)

This section reports first estimates of  $c_4$ , which controls how the monopile-induced TKE source affects the dissipation equation.

### Step 11: Estimate $c_4$

The parameter  $c_4$  appears in  $S^{pile\_e} = c_4 (\epsilon/k) S^{pile\_k}$ . A direct estimate would require the LES-equivalent source term in the  $\epsilon$ -equation, which is difficult to diagnose. Therefore, two approaches are proposed.

First, use a local-equilibrium estimate:

$$c_4 \approx C\epsilon^2 \times \langle \epsilon_{LES} \rangle_{Vc} / \langle S^{pile\_k,LES} \rangle_{Vc}$$

Second, use offline calibration. Run a simplified  $k$ - $\epsilon$  model with the LES-derived  $S^{pile\_k}$ , test different values of  $c_4$ , and choose the value that best reproduces the LES-derived coarse-grained profiles of  $k$ ,  $\epsilon$ ,  $l$ , where the turbulence length scale is

$$l = C\mu^{(3/4)} \times k^{(3/2)} / \epsilon$$

The offline calibration approach is recommended because  $c_4$  is a model-closure parameter rather than a directly measured LES variable.

### 8.1 Local-equilibrium estimate

*Note: This estimate should be treated cautiously: the near wake of a monopile may not satisfy local equilibrium.*

**Using 600\*120\*40 volume**

| Field                                                | Value                                                     | Unit      |
|------------------------------------------------------|-----------------------------------------------------------|-----------|
| $\epsilon^{LES}$ (volume-averaged)                   | $4.77292 / 2,880,000 = 1.657 \times 10^{-6}$<br>$m^2/s^3$ | $m^2/s^3$ |
| $\langle S^{ks3es}_k, LES \rangle$ (volume-averaged) | $2.153 \times 10^{-6}$                                    | $m^2/s^3$ |
| $c_4$ (local-equilibrium estimate)                   | $1.92 \times 1.657e-6 / 2.153e-6 = 1.478$                 |           |

**Using 300\*100\*40 volume**

| Field                                                | Value                                                     | Unit      |
|------------------------------------------------------|-----------------------------------------------------------|-----------|
| $\epsilon^{LES}$ (volume-averaged)                   | $4.51117 / 1,200,000 = 3.759 \times 10^{-6}$<br>$m^2/s^3$ | $m^2/s^3$ |
| $\langle S^{ks3es}_k, LES \rangle$ (volume-averaged) | $5.95977 / 1,200,000 = 4.966 \times 10^{-6}$              | $m^2/s^3$ |
| $c_4$ (local-equilibrium estimate)                   | $1.92 \times 3.759e-6 / 4.966e-6 = 1.453$                 |           |

## 9. Summary of Key Results

| Parameter                                           | Value                  | Notes                             |
|-----------------------------------------------------|------------------------|-----------------------------------|
| $C_{D,eff}$                                         | 0.404                  | From drag force on cylinder       |
| $\langle k_{res} \rangle_{Vc}$                      | $2.898 \times 10^{-4}$ | Resolved turbulent kinetic energy |
| $P_{D^{Rak}} \text{ (m}^2/\text{s}^3\text{)}$       | $4.83 \times 10^{-6}$  | Drag power per unit mass          |
| $S_{k_{sgs}}^{LES} \text{ (m}^2/\text{s}^3\text{)}$ | $2.15 \times 10^{-6}$  | Volume-averaged TKE source        |
| $\alpha^k$                                          | 0.445                  | TKE fraction of drag power        |
| $c_4$                                               | 1.478                  | Dissipation equation parameter    |
| $\epsilon^{LES} \text{ (m}^2/\text{s}^3\text{)}$    | $1.66 \times 10^{-6}$  | Volume-averaged dissipation       |

## 10. Discussion

### 10.1 Comparison to Published Values

This section compares the  $\alpha^k$  and  $c_4$  estimates from this study against the five most relevant published studies, selected for having flow conditions closest to  $Re = 4 \times 10^6$ .

| Study                      | Method / Estimated Re             | Re                                               | $\alpha^k$ | $c_4$        | Notes                                                                    |
|----------------------------|-----------------------------------|--------------------------------------------------|------------|--------------|--------------------------------------------------------------------------|
| This study                 | LES                               | $= 4 \times 10^6$                                | 0.445      | 1.478        | Control volume 600×120×40 m                                              |
| Rennau et al. (2012)       | RANS closure, Baltic Sea field Re | —                                                | —          | 0.6–1.4      | Source of the $c_4/C_d$ closure; range spans strong to weak mixing cases |
| Christiansen et al. (2023) | SCHISM, German Bight field Re     | $\approx 1.6 \times 10^6$ –<br>$3.2 \times 10^6$ | —          | 1.0 (reused) | Reuses Rennau closure at real tidal current speeds                       |

### 10.2 Physical interpretation of $\alpha^k$

| Question                                                           | Answer                                                                                                                          |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| What fraction of drag power becomes TKE?                           | About 44.5%. Remainder likely lost to advection, pressure work, or unresolved SGS effects.                                      |
| How does this compare to the common assumption of $\alpha^k = 1$ ? | The actual value 0.445 is roughly half the commonly assumed value, so $\alpha^k = 1$ would overpredict monopile-induced mixing. |

## 10.3 Physical interpretation of $c_4$

| Question                                                              | Answer                                                                                                                                                                                                   |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| How does $c_4$ relate to standard $k$ - $\epsilon$ closure constants? | $c_4$ (1.478) is below $C\epsilon_2$ (1.92) but close in magnitude, meaning the monopile-induced dissipation source is comparable in scale to standard shear-driven dissipation, not a minor correction. |

## 10.4 Limitations and assumptions

| Limitation                                      | Impact / Notes                                                                                                                                                                                                                                                     |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SGS terms included in $\epsilon$ but not in $k$ | $\epsilon_{LES}$ includes $\epsilon_{SGS}$ via $\nu_{t,SGS}$ , but $k_{res}$ excludes SGS kinetic energy. This asymmetry means the $\epsilon/k$ ratio used for $c_4$ mixes a complete dissipation estimate with an incomplete $k$ estimate, which may bias $c_4$ . |
| Local equilibrium assumption for $c_4$          | $c_4 = 1.478$ assumes production and dissipation are locally balanced in the control volume. This is uncertain in the near wake, where advection and transport are significant, so the value should be treated as a rough first estimate.                          |
| Offline calibration not yet performed           | The instructions recommend calibrating $c_4$ by running a simplified $k$ - $\epsilon$ model against LES $k$ , $\epsilon$ , and length-scale profiles. This has not been done, so no second, more reliable $c_4$ estimate exists yet.                               |

## 11. Final Ocean-Model Parameterization

The LES-informed parameterization suitable for implementation in a regional ocean model is:

$$S_{ks\epsilon s\_k} = \alpha^k \times \frac{1}{2} \times C_{D,eff} \times (D / A_{CEL}) \times |u_h|^3$$

$$S_{ks\epsilon s\_ \epsilon} = c_4 \times (\epsilon/k) \times S_{ks\epsilon s\_k}$$

| Parameter                | Value                                             |
|--------------------------|---------------------------------------------------|
| $\alpha^k$ (estimated)   | 0.445                                             |
| $c_4$ (estimated)        | 1.478                                             |
| $C_{D,eff}$ (estimated)  | 0.404                                             |
| Control volume size used | 600 x 120 x 40 m                                  |
| Averaging period         | 32 s                                              |
| SGS terms                | Included in $\epsilon$ via nut; excluded from $k$ |